

ASSIGNMENT-7

$$\textcircled{1} \quad h = y_{i+1} - y_i = 2 - 0 = 2. \quad \text{all } h = 2$$

$$f_0 v_0 = 0,$$

$$H_1 v_1 = 0.1$$

$$H_2 v_2 = 1.5 \times 0.12 = 0.18$$

$$H_3 v_3 = 3 \times 0.2 = 0.6$$

$$H_4 v_4 = 3.5 \times 0.25 = 0.875$$

$$H_5 v_5 = 3.2 \times 0.3 = 0.96$$

$$H_6 v_6 = 2 \times 0.15 = 0.3$$

$$H_7 v_7 = 0$$

$\textcircled{2}$

$$I = \frac{h}{2} [f_0 + 2(f_1 + f_2 + f_3 + f_4 + f_5 + f_6) + f_7]$$

$$A = \frac{2}{2} [H_0 + 2(H_1 + H_2 + H_3 + H_4 + H_5 + H_6) + H_7]$$

$$A = 1 \times [0 + 2(1 + 1.5 + 3 + 3.5 + 3.2 + 2) + 0]$$

$$A = 1 [0 + 2(14.2) + 0] = 28.4 \text{ m}^2$$

$$\textcircled{2} Q = \frac{2}{2} [H_{U_0} + 2(H_{U_1} + H_{U_2} + H_{U_3} + H_{U_4} + H_{U_5} + H_{U_6}) + H_{U_7}]$$

$$Q = 1 [0 + 2(0.1 + 0.18 + 0.6 + 0.875 + 0.96 + 0.3) + 0]$$

$$Q = 1 [0 + 2(3.015) + 0] =$$

$$Q = 6.03 \text{ m}^3/\text{sec}$$

$$\textcircled{5} \bar{v} = \frac{Q}{A} = \frac{6.03}{28.4} = 0.2123 \text{ m}^3/\text{sec}$$

$$\textcircled{b} 3k_1 + 1 + 2k_2 = 8$$

$$3k_1 + 2k_2 = 7$$

($k_1 = 1, k_2 = 2$) only sol. ^{int.} and, we can

make 4 elements (0 to 4)

4 elem (4 to 7)

Basically. $\frac{1}{3}$ Rule for 4 interval (0 to 4)

$\frac{3}{8}$ Rule for 3 intervals (4 to 7)

$$\textcircled{1} \quad A_{1/3} = \frac{h}{3} [H_0 + 4(H_1 + H_3) + 2H_2 + H_4]$$

$$A_{1/3} = \frac{2}{3} [0 + 4(1+3) + 2(1.5) + 3.5]$$

$$A_{1/3} = \frac{2}{3} [0 + (6+3+3.5)] = 15.$$

Points 4 to 7:

$$A_{3/8} = \frac{3h}{8} [H_4 + 3(H_5 + H_6) + H_7]$$

$$A_{3/8} = \frac{3(1)}{8} [3.5 + 3(3.2+2) + 0]$$

$$A_{3/8} = 0.375 [3.5 + 3(5.2)] = 0.375 \times 19.1 \\ = 14.325$$

$$\text{Total Area } A = A_{1/3} + A_{3/8} = 15 + 14.325 \\ = 29.325 \text{ m}^2$$

② Discharge (Q):-

$$Q_{1/3} = \frac{h}{3} [Hv_0 + 4(Hv_1 + Hv_3) + 2Hv_2 + Hv_4]$$

$$Q_{1/3} = \frac{2}{3} [0 + 4(0.1 + 0.5) + 2(0.8) + 0.875]$$

$$Q_{1/3} = \frac{2}{3} [0 + 4(0.6) + 0.36 + 0.875]$$

$$= \frac{2}{3} [4.035] = 2.69.$$

3/8 rule:

$$Q_{3/8} = \frac{2h}{8} [Hv_4 + 3(Hv_5 + Hv_6) + Hv_7]$$

$$Q_{3/8} = \frac{6}{8} [0.875 + 3(0.96 + 0.3) + 0]$$

$$Q_{3/8} = 0.75 [0.875 + 3(1.26)]$$

$$= 0.75 \times 4.655 = 3.49125$$

$$Q = Q_{1/3} + Q_{3/8} = 2.69 + 3.49125 = 6.18125 \text{ m}^3/\text{s}$$

$$\bar{U} = Q/A = 6.18125 / 29.325 = 0.2108 \text{ m/sec}$$

Method	A	Δ	$\bar{U}$
①			
Trapezoidal	28.40	6.030	0.2122
Simpson's	29.325	6.181	0.2108.

② $f(t) = (8 + 5\cos^2(0.4t))(5e^{-0.5t} + 3e^{0.15t})$

$$f(2) = 61.4043$$

$$f(4) = 130.5038$$

$$f(5) = 59.9461$$

$$f(3.5) = 48.2800$$

$$f(6.5) = 95.0904$$

$$f(2.75) = 52.3303$$

(Directly entered into
calc w/only results)

$$f(4.25) = 50.6999$$

$$f(5.75) = 75.5159$$

$$f(7.25) = 114.8538$$

$$\rightarrow \textcircled{1} \quad i=1, h=6$$

$$R_{11} = \frac{6}{2} [f(2) + f(8)] = 3(61.4643 + 130.5025) \\ = 575.7244$$

$$\textcircled{2} \quad i=2, h=3$$

$$R_{21} = 0.5 R_{11} + 3[f(5)] = 0.5(575.7244) \\ + 3(59.4461) \\ = 467.7004$$

$$R_{22} = R_{21} + \frac{R_{21} - R_{11}}{3} = 467.7004 \\ + 467.7004 - \frac{575.7244}{3} \\ = 431.6924$$

$$\textcircled{3} \quad i=3, h=1.5$$

$$\begin{aligned} R_{31} &= 0.5 R_{21} + 1.5 [f(3.5) + f(6.5)] \\ &= 0.5 (467.7004) + 1.5 [48.3800 + 95.0904] \\ &= 449.0557 \end{aligned}$$

$$\begin{aligned} R_{32} &= R_{31} + \frac{R_{31} - R_{21}}{3} = 449.0557 \\ &\quad + \frac{449.0557 - 467.7004}{3} \\ &= 442.8409 \end{aligned}$$

$$\begin{aligned} R_{33} &= R_{32} + \frac{R_{32} - R_{22}}{1.5} = 442.8409 \\ &\quad + \frac{442.8409 - 431.6924}{1.5} \\ &= 443.5841 \end{aligned}$$

$$\textcircled{4} \quad i=4, h=0.75$$

$$R_{41} = 0.5R_{31} + 0.75[f(2.75) + f(4.25) + f(5.75) + f(7.25)]$$

$$0.5(449.0557) + 0.75(52.303 + 50.6999 + 75.5159 + 114.8538) \\ = 444.5778$$

$$R_{42} = R_{41} + \frac{R_{41} - R_{31}}{3} \\ = 444.5778 + \frac{444.5778 - 449.0557}{3} \\ = 443.0851$$

$$R_{43} = R_{42} + \frac{R_{42} - R_{32}}{15} = 443.0851 + \frac{443.0851 - 442.8409}{15} \\ = 443.1014.$$

$$R_{44} = R_{43} + \frac{R_{43} - R_{33}}{63} = 443.1014 + \frac{443.1014 - 443.5841}{63}$$

$$= 443.0937$$

$$\varepsilon_q = \left(\frac{R_{q4} - R_{q3}}{R_{q4}} \right) \times 100 = \left(\frac{443.0937 - 443.1014}{443.0937} \right) \times 100$$

$$= 0.0017\%$$

$\therefore 0.0017\% < 0.1\%$ we can accept

$$\boxed{443.0937 \text{ mg}}$$

from code:

$$final = 443.09372415683464$$

```
PS D:\ES0208\Computational_Methods_In_Engineering\Assignment_7\q2> python .\q2.py
Final=443.09372415683464
```

①

```
PS D:\ES0208\Computational_Methods_In_Engineering\Assignment_7> python .\q1.py
Trapezoidal: A=28.4, Q=6.029999999999999, U=0.2123
Simpson: A=29.325000000000003, Q=6.18125, U=0.2108
```

